

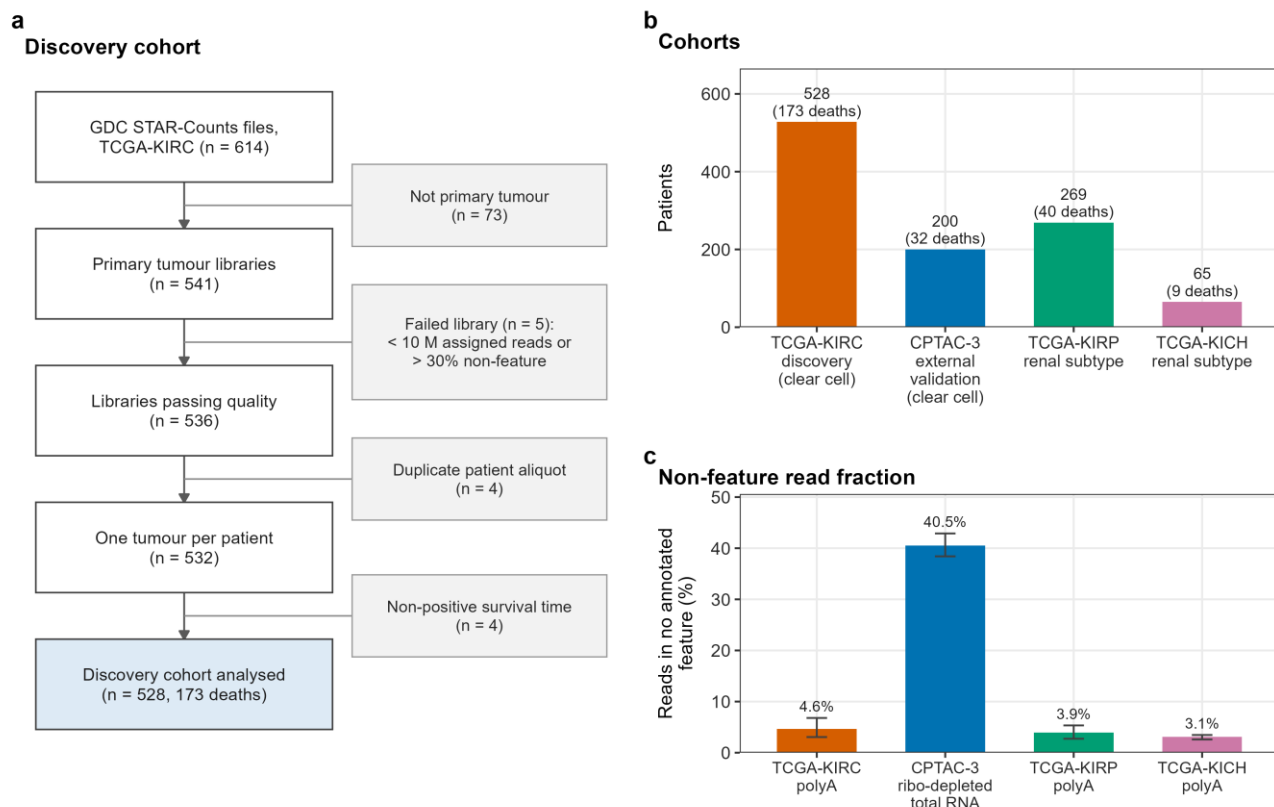


Fig. 1. Study design, cohorts and library protocol. **a** Assembly of the discovery cohort. Of 614 TCGA-KIRC STAR-Counts files downloaded from the Genomic Data Commons, 73 were not from primary tumours, leaving 541 primary-tumour libraries. Five failed the library rule (fewer than 10 million assigned reads or more than 30% non-feature reads), leaving 536. Four were duplicate aliquots, leaving 532 tumours from distinct patients, and 4 patients had non-positive or missing survival time. The shaded box is the analysed discovery cohort of 528 patients with 173 deaths. **b** Patients per cohort, with deaths in parentheses: TCGA-KIRC 528 (173), CPTAC-3 200 (32), TCGA-KIRP 269 (40) and TCGA-KICH 65 (9), totalling 1,062 patients and 254 deaths. The TCGA-KIRP bar counts the 40 deaths among the 263 patients with a recorded T category, who enter the adjusted models. Of all 269 TCGA-KIRP patients with module scores, 42 died. Colours identify cohorts, as in **c**. **c** Median non-feature read fraction per cohort, the percentage of reads in the STAR gene-counts summary assigned to no annotated feature. The denominator sums the unmapped, multimapping, non-feature, ambiguous and gene-assigned counts. Bars are medians and whiskers interquartile ranges: TCGA-KIRC 4.6% (3.1 to 6.8), CPTAC-3 40.5% (38.4 to 42.9), TCGA-KIRP 3.9% (2.7 to 5.3) and TCGA-KICH 3.1% (2.6 to 3.5), on 528, 200, 269 and 65 libraries. Library protocols are given under the cohort labels. The ribo-depleted total-RNA cohort is about 9 to 13 times higher than the poly(A) cohorts (8.7-fold against TCGA-KIRC and 13.0-fold against TCGA-KICH). Each cohort used one protocol, so protocol and cohort cannot be separated.

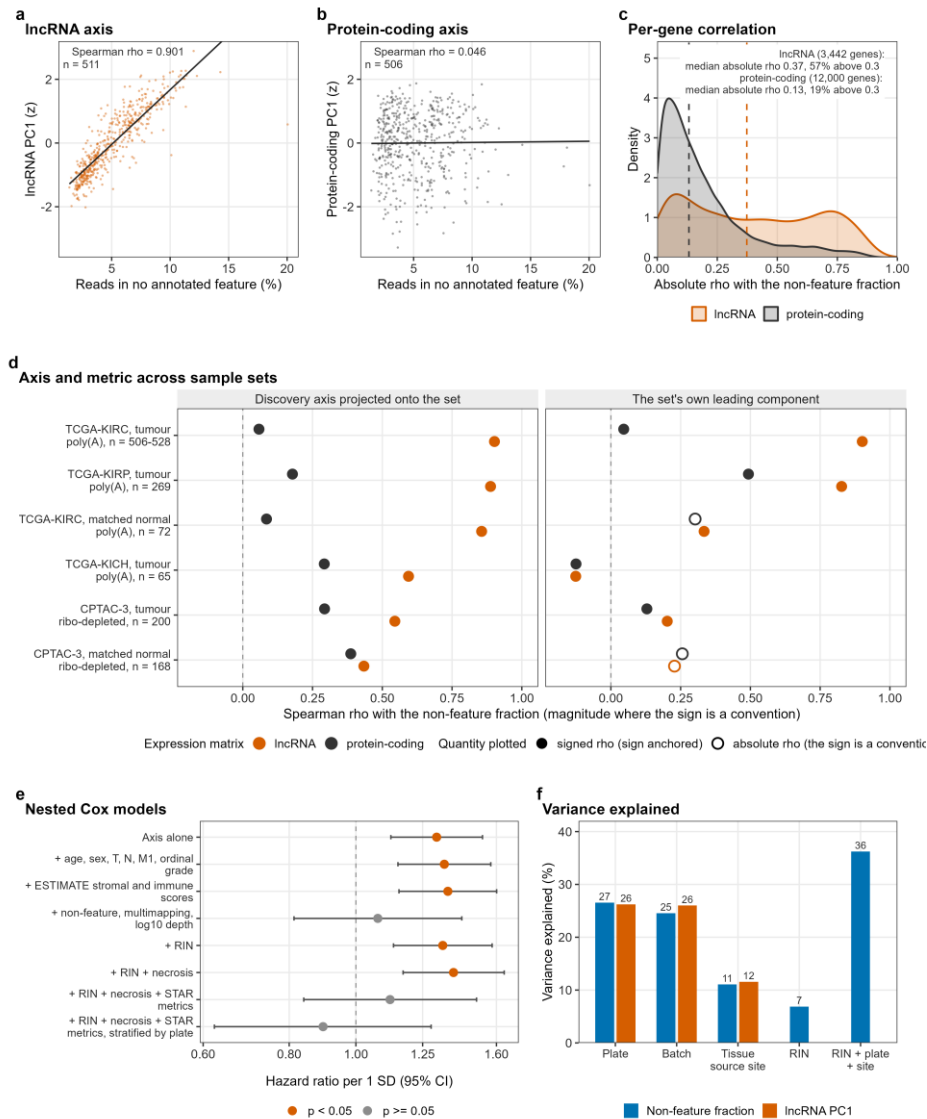


Fig. 2. The leading lncRNA expression axis tracks non-exonic read content. **a** First principal component of the observed lncRNA matrix, $\log_2(\text{FPKM} + 1)$, against the non-feature read fraction per library, with a least-squares fit (Spearman rho 0.90, $n = 511$). **b** As **a** for the protein-coding matrix, on identical axes (rho 0.05, $n = 506$). The observed protein-coding red eigengene (473 genes, RNA splicing) does track the metric (rho 0.84, Supplementary Fig. 5a). **c** Densities of per-gene absolute Spearman correlation with the metric. Dashed medians are 0.37 for 3,442 lncRNAs (56.8% above 0.3) and 0.13 for 12,000 protein-coding genes (18.8%). **d** Spearman correlation of the axis with the metric in six sample sets, faceted by the projected discovery axis and each set's own leading component. Open symbols, plotted as magnitudes, mark the 3 of 12 own-component rows whose sign cannot be fixed because its correlations with mean expression and the projected axis are weak or disagree. The projected lncRNA axis gives rho 0.90, 0.89, 0.86, 0.59, 0.55 and 0.43 in TCGA-KIRC tumours ($n = 528$), TCGA-KIRP (269), TCGA-KIRC normal kidney (72), TCGA-KICH (65), CPTAC-3 tumours (200) and CPTAC-3 normal tissue (168). Within each set the projected lncRNA correlation exceeds the projected protein-coding correlation by 0.30 to 0.84 in the four poly(A) sets and by 0.05 to 0.25 in the two ribo-depleted sets. **e** Hazard ratio per standard deviation of the lncRNA axis under eight nested Cox models (503 patients, 163 events), with 95% confidence intervals on a logarithmic axis. Red marks $p < 0.05$. The hazard ratio is 1.31 to 1.39 before the three STAR metrics enter and 1.08 after. It is 1.12 with RNA integrity and slide necrosis added to the STAR metrics, and 0.90 with plate stratification. **f** Variance in the metric and the axis explained by plate (26.6% and 26.2%), batch (24.6% and 26.0%) and tissue source site (11.1% and 11.6%), as one-way analysis of variance R-squared on 511 to 528 libraries. Two further bars, for the metric only, give regression R-squared on the 527 patients with an integrity number, for RNA integrity alone (6.9%, 1 degree of freedom) and for RNA integrity with plate and site (36.2%, 25 degrees of freedom). The joint bar comes from a single model, so it need not equal the sum of the other bars.

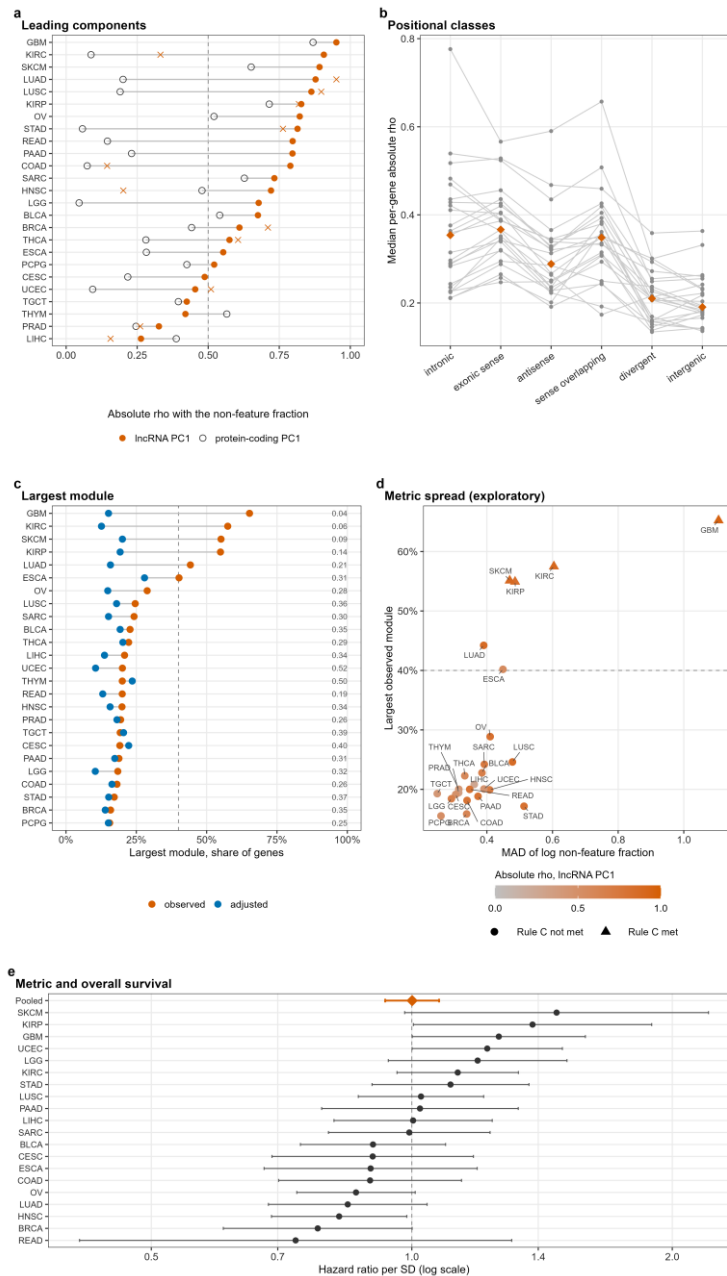


Fig. 3. The lncRNA axis and network reorganisation across TCGA. Twenty-five cancer types with at least 100 retained primary tumours. Dashed lines are the decision rules, fixed before any non-renal project was analysed (Methods). **a** Absolute Spearman correlation with the non-feature read fraction of the leading principal component of the lncRNA (filled) and protein-coding (open) matrices, and of the lncRNA component in solid-tissue normals with at least 20 libraries (crosses). Rule A, for a lncRNA-specific axis (absolute rho at least 0.5 and at least 0.3 above protein-coding), is met in 9 of 25, 8 of them non-renal. **b** Median per-gene absolute correlation with the metric by positional lncRNA class, one line per cancer type, with diamonds for the median over types. Intronic exceed intergenic lncRNAs in all 25. **c** Share of lncRNAs in the largest module of the observed (orange) and adjusted (blue) networks, with the adjusted Rand index between the two partitions at right. Rule C, for a reorganised network, is met in GBM, KIRC, SKCM and KIRP. **d** Exploratory panel, specified after the locked rules were applied. Within-cohort spread of the metric (median absolute deviation of its logarithm) against the share of genes in the largest observed module, coloured by the absolute rho of the lncRNA component. Triangles meet Rule C. Spread correlates with the largest-module share at Spearman rho 0.69 (false discovery rate 0.0016) and with the absolute rho of the lncRNA component at 0.73 (6.1e-04). **e** Hazard ratio for overall survival per standard deviation of the non-feature fraction in the 20 cancer types with at least 20 deaths, adjusted for age, sex and stage where recorded and stratified by sequencing plate. Error bars are 95% confidence intervals, and the diamond is the random-effects pooled estimate, 1.00 (0.93 to 1.08, I-squared 50%). Cancer types are TCGA project codes. Source data are in Supplementary Table 6.

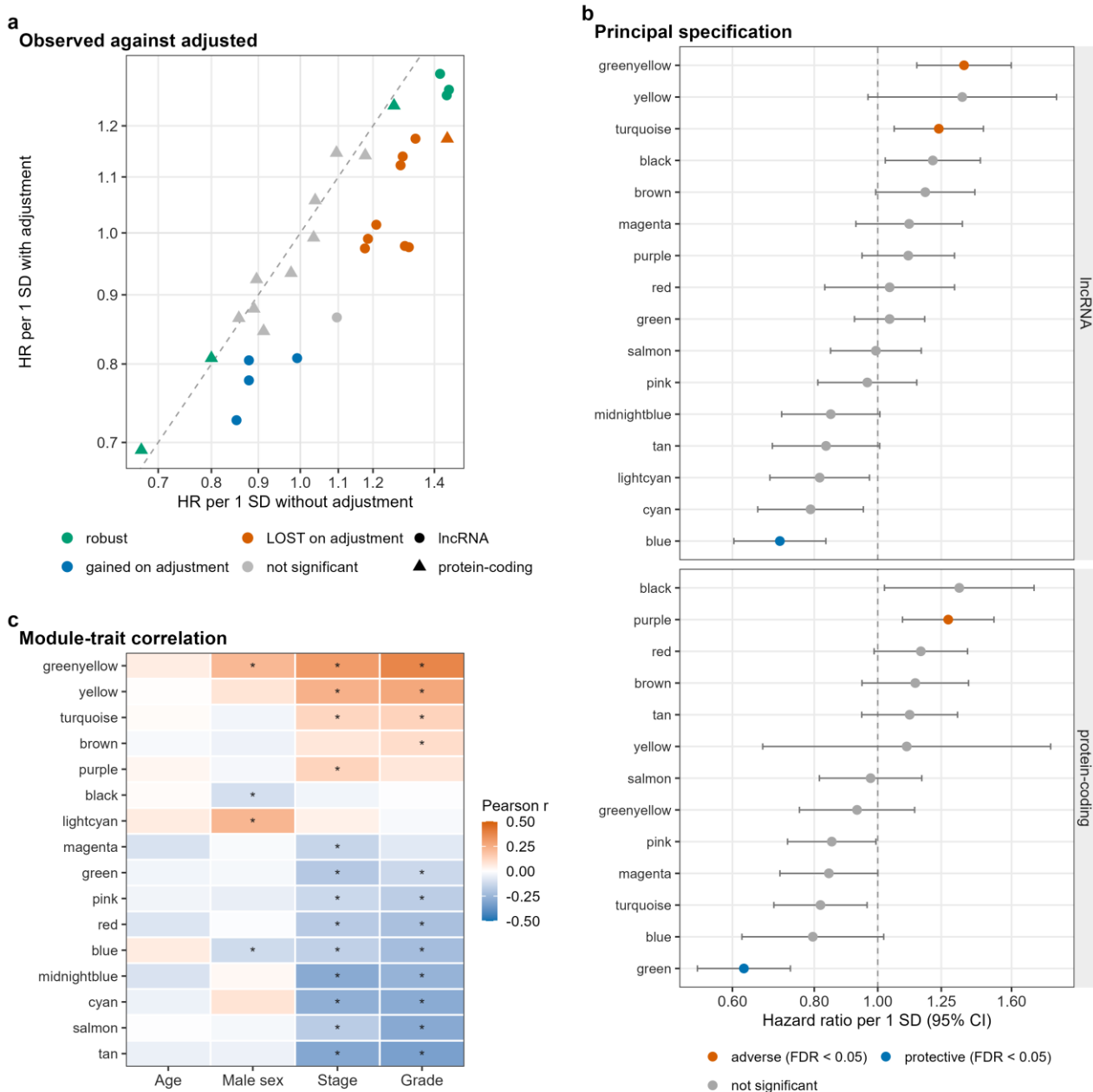


Fig. 4. Adjusting for the alignment metrics changes which co-expression modules are prognostic. **a** Hazard ratio per standard deviation of each of the 29 module eigengenes, computed on the observed matrix (horizontal axis) and on the matrix residualised on the three STAR alignment metrics (vertical axis). Module membership is the same on both axes, so only the adjustment differs. Axes are logarithmic and the dashed line marks equality. Colour shows the outcome of adjustment and shape the network. Of 29 modules, 15 were significant at a 5% false discovery rate without adjustment and 10 with it: 9 lost significance, 4 gained it, 6 were significant in both arms and 10 in neither. Models adjust for age, sex, T category, nodal status, M1 and ordinal grade and are fitted on 504 patients with 163 events (lncRNA network) and 502 with 165 (protein-coding network). False discovery rates are controlled within network and arm. **b** Hazard ratio per standard deviation of each module eigengene under the principal specification, by network, with 95% confidence intervals on a logarithmic axis. The principal specification adds the eigengene to age, sex, T category, nodal status, M1, ordinal grade, the ESTIMATE stromal and immune scores and the three STAR metrics, in 504 patients with 163 events (lncRNA network) and 502 with 165 (protein-coding network). Colour marks the direction of modules with a within-network false discovery rate below 0.05, and grey marks non-significant modules. Five of 29 modules meet that threshold: protein-coding green and purple, and lncRNA blue, greenyellow and turquoise. **c** Pearson correlation of the 16 lncRNA module eigengenes with age, male sex, ordinal AJCC stage and ordinal grade in 511 patients. Asterisks mark a Benjamini-Hochberg false discovery rate below 0.05 across the panel's 64 tests (29 of 64).

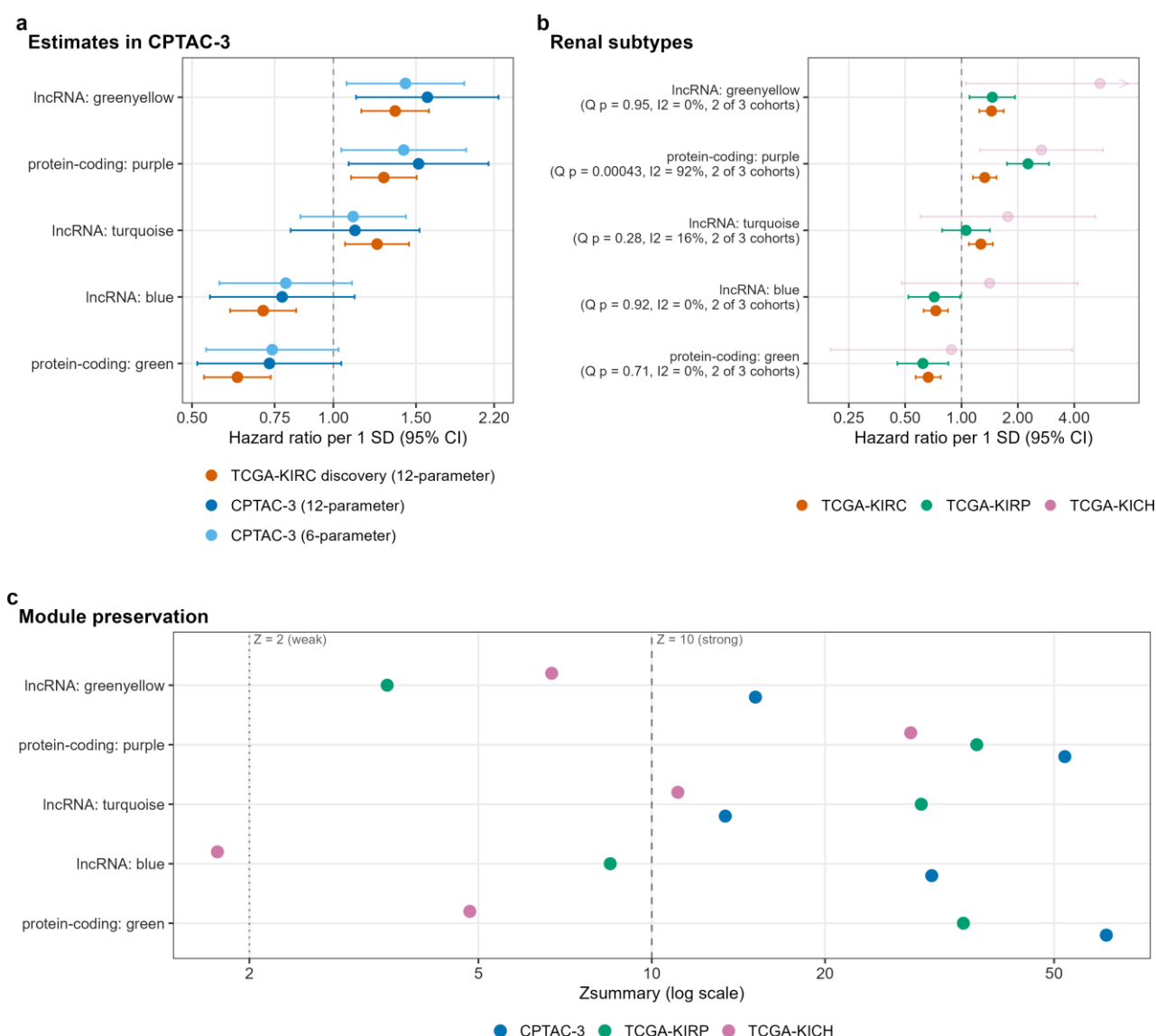


Fig. 5. External association estimates, subtype estimates and module preservation. **a** Three estimates for each of the five modules significant in discovery. These are TCGA-KIRC under the principal specification (504 patients and 163 events for lncRNA modules, 502 and 165 for protein-coding modules), CPTAC-3 with the same twelve-parameter covariate set, and CPTAC-3 with a six-parameter set of module eigengene, age, sex, T category, M1 and ordinal grade. Both CPTAC-3 arms use 199 patients with 32 events, with membership and loadings fixed in discovery and only the module coefficient re-estimated. Points are hazard ratios per standard deviation with 95% confidence intervals on a logarithmic axis. All five keep the discovery direction in both arms (one-sided sign test $p = 0.031$), and greenyellow and purple are nominally significant (p below 0.05) in both. Replication was defined as a false discovery rate below 0.05 in the six-parameter arm (5.3 events per parameter), and no module meets it. The twelve-parameter arm, in which greenyellow and purple reach a false discovery rate of 0.042, has 2.7 events per parameter, below the threshold of 5 used elsewhere, and is not used to claim replication. **b** The same five modules projected with the fixed discovery loadings onto TCGA-KIRC (497 patients, 160 events), TCGA-KIRP (263, 40) and TCGA-KICH (65, 9), adjusted for one parsimonious set of age, sex, T category, M1 and the non-feature fraction. Error bars are 95% confidence intervals on a logarithmic axis. Row labels give Cochran Q p values and I-squared for TCGA-KIRC and TCGA-KIRP, the two cohorts meeting the events-per-parameter rule. TCGA-KICH, at 1.5 events per parameter, is faded and descriptive only. Its lncRNA greenyellow interval extends beyond the axis limit and is marked. **c** Preservation of the five modules in CPTAC-3 ($n = 200$), TCGA-KIRP (269) and TCGA-KICH (65), as WGCNA Zsummary on a logarithmic axis, from 100 permutations for the lncRNA network and 30 for the protein-coding network. Reference lines mark the conventional weak ($Z = 2$) and strong ($Z = 10$) thresholds, and 10 of the 15 module-cohort pairs lie above 10.

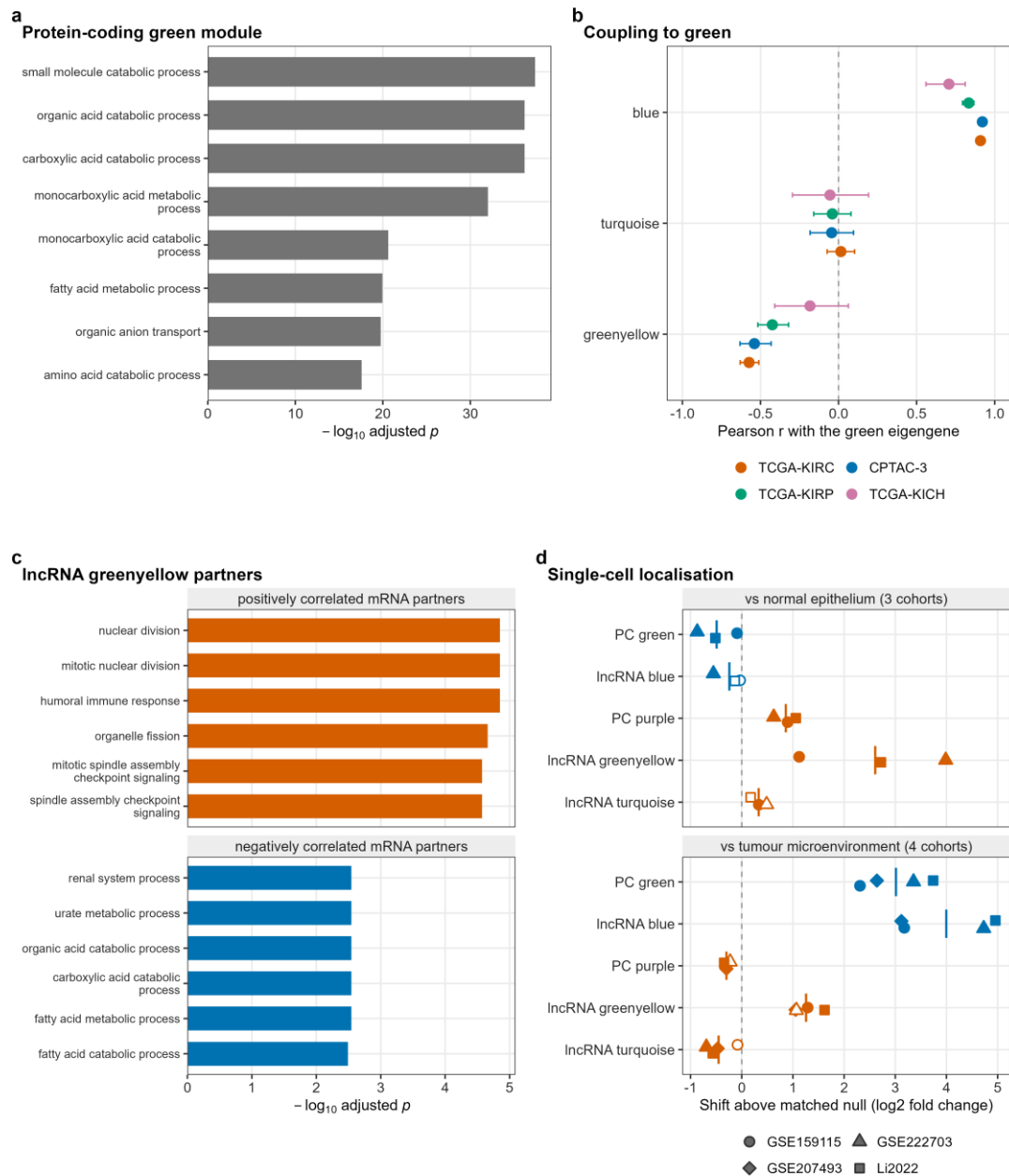


Fig. 6. What the prognostic modules are coupled to and where they are expressed. **a** The eight Gene Ontology biological-process terms with the smallest adjusted p values for the protein-coding green module, against the 12,000 protein-coding genes entering the network. Bars are $-\log_{10}$ Benjamini-Hochberg adjusted p. Under the principal specification the module's hazard ratio is 0.62 per standard deviation (95% confidence interval 0.53 to 0.74, false discovery rate $2.2e-07$). **b** Pearson correlation, with 95% confidence intervals, of each prognostic lncRNA module eigengene with the protein-coding green eigengene in 497 TCGA-KIRC, 199 CPTAC-3, 269 TCGA-KIRP and 65 TCGA-KICH patients. Two of three intervals exclude zero in TCGA-KIRC, CPTAC-3 and TCGA-KIRP, and one of three in TCGA-KICH. The turquoise interval spans zero in every cohort. **c** Gene Ontology terms of the protein-coding genes whose expression tracks the lncRNA greenyellow eigengene. The 300 most strongly correlated partners of each sign are tested separately, and the six terms with the smallest adjusted p are shown per sign. The module's hazard ratio is 1.35 (1.15 to 1.60, false discovery rate 0.0027). Positively correlated partners are enriched for a proliferation and humoral-immune programme, and negatively correlated partners for the catabolic and renal-tubular programme of **a**. **d** Single-cell localisation of the five prognostic modules. For each module and cohort, the median log2 fold change of members in malignant cells against normal kidney epithelium (GSE159115, GSE222703 and Li2022) and against pooled tumour immune, endothelial and stromal cells (the same three and GSE207493), minus the mean of 2,000 abundance-matched random gene sets of the same gene class. Shape marks the cohort, colour the hazard direction, filled symbols an empirical p below 0.05, and vertical bars the mean over cohorts. Members tested per cohort range from 14 to 18 for greenyellow against normal epithelium and are listed with all values in Supplementary Table 19.

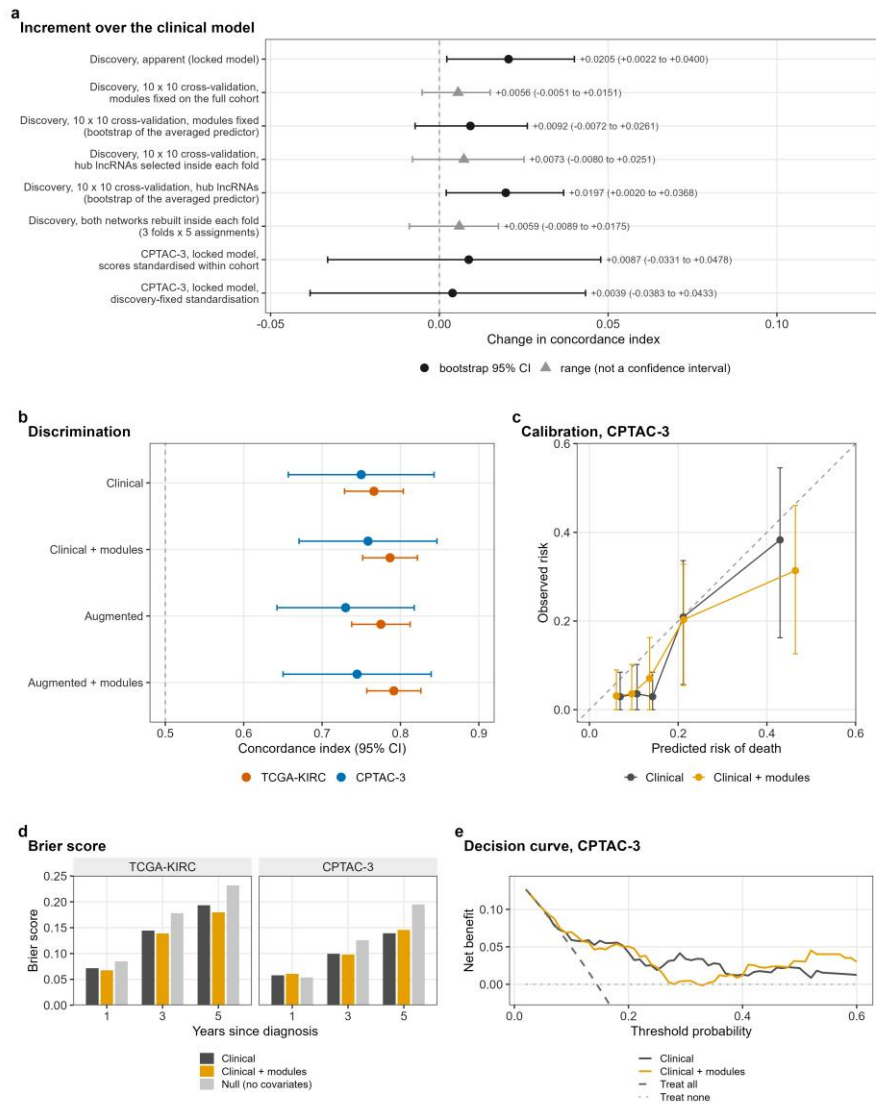


Fig. 7. Module eigengenes add no demonstrable discrimination over the clinical model. All panels are for overall survival. The clinical model is age, sex, T category, nodal status, M1 and ordinal grade. The augmented model adds the ESTIMATE stromal and immune scores and the three STAR metrics. Module eigengenes enter a penalised Cox model with the comparator terms unpenalised (Table 3). **a** Increment in concordance index over the clinical model under six estimation schemes (eight rows), in 493 discovery patients with 159 events and 199 CPTAC-3 patients with 32. Black circles carry paired patient bootstrap 95% confidence intervals (2,000 resamples), and grey triangles carry ranges across repeats, which are not confidence intervals. Row one is the locked model scored in sample (+0.0205). Rows two to five are ten repeats of ten-fold cross-validation with module eigengenes (+0.0092) and with hub lncRNAs selected inside each fold (+0.0197), each range row preceding its bootstrap row. Row six rebuilds both networks inside each training fold, over three folds and five assignments (+0.0059). The last two rows are the locked model in CPTAC-3 under cohort (+0.0087) and discovery-fixed (+0.0039) standardisation. **b** Concordance index with 95% confidence intervals for four locked models in both cohorts (493 patients with 159 events in TCGA-KIRC, 199 with 32 in CPTAC-3), eigengenes standardised within cohort. The dashed line marks 0.5, and TCGA-KIRC values are apparent (in-sample). **c** Three-year calibration in CPTAC-3 (199 patients, 32 events) by quintile of predicted risk, with the baseline hazard fixed in discovery. The dashed line is perfect calibration, and error bars are Kaplan-Meier 95% confidence intervals. The calibration slope is 0.97 (0.60 to 1.34) for the clinical model and 0.89 (0.58 to 1.20) with modules, and observed over expected risk is 0.75 and 0.74. **d** Brier scores at one, three and five years, weighted by the inverse probability of censoring (lower is better), for the clinical model, the model with modules and a null model assigning every patient the Kaplan-Meier risk. Modules improve the score at 4 of 6 cohort-by-horizon combinations, three of them apparent TCGA-KIRC scores, and at 1 of 3 horizons in CPTAC-3. **e** Three-year decision curve in CPTAC-3 with the same fixed baseline, thresholds 0.02 to 0.60. The median change in net benefit over thresholds 0.10 to 0.40 is -0.004, and the change is positive at 9 of 31 thresholds. The treat-all curve lies below the plotted range.